ActInf Textbook Group ~ Cohort 7 ~ Session 5 (Chapter 2 +++ Open Discussion) ~ 8/12/2024

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Duration: 47:02 | **Uploaded**: Unknown **Transcript method**: auto_caption

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all right we're in an open slen session so Adam perhaps begin with the question and then Andrew go for it okay thank you so in chapter seven in the summary it says that um inference using both simple and more complex generative models can always proceed through free energy minimization which illustrates the generality of the approach and my question is about uh whether or not there is a partition function for the model of the minimum free energy State uh and I wanted to note how about in section 7.3 they talk about sorting the states of the world into categories of whether the could treater has control or whether it does not and then it says that the the transition prob ilities for the ones that it doesn't are invariant and then each state that the creature has a control over the world has like a unique transition probability so then it's always going to try to select the transition probability which is most effective at bringing it towards a free energy minimum state right so um that would be the equilibrium I suppose which was making me think about um whether or not there's a partition function yeah so uh it's kind of like to answer that there are a few different pieces going on so yeah with like you can see in that chapter where um let's see page 126 the the just the second page of the chapter you can see like like the breakdown of the the B Matrix as they call it at the bottom of the page and so like yeah you're going to have a unique B Matrix for uh every hidden State Factor um and it's the dimensionality is kind of like um the we're gonna what is the state going to be at the next time step conditioned on uh what you infer the state to be in the current time step as well as um a a policy so it's kind of like the agent will uh you know they'll try to think what can I do now to change what I think is going on now in terms of the Hidden state to um you know a particular hidden state that I I kind of want to see in the next step and I I would say in a kind of all things controlled for kind of way what you said is correct it it will you know go with what the action that is you know most probable uh to get it to a state it wants to be in and that said because free energy is being run across all of these different you know components of the model um it's not necessarily the case that you can just look at the B Matrix and immediately know what the agent is going to do just because one state trans position is more probable than another um when it's Computing expected free energy for example that you know gets broken down into epistemic and pragmatic value um an agent might choose to transition it might choose to do something to make a state transition that it isn't really clear on for example it might compute like oh there's epistemic value there um you know and then if you have your agent like learning and B Matrix those values will get updated over time um so I I hope that that kind of framing is is useful in some way as far as a partition function um so I just want to I want to better understand and maybe it'll be helpful for others too how how would you define a partition function in in the way that you mean it yeah you know what um I don't even really know like honestly I've never actually formally studied um statistical physics so when I like I've asked there's also like a partition function in is it number Theory and I was I asked all over nil about it on on YouTube and he gave me this I'm gonna be rambling on he he sent me a paper that was actually discussing like the relationship between the partition function in number Theory with the partition function in statistical physics in my question to him was was it the same one and he said he didn't really know and I had the paper I was reading it was interesting I forget I don't have it off hand and um in this case I'm kind of wondering like the word partitioning in terms of sorting the two states of the world from the ones we can change the ones that we cannot and then at equilibrium like the um those States become stationary I suppose in some sense and uh so but I'm also wondering I'm also wondering is that would the partition function manifest as the different aspects of active inference that are supposedly emerging like let me give a few few thoughts on this just really briefly because you're you're in the right track partitions being used in a um hate to reuse the pun but it's a particular way and so a lot of the momentum in the current kind of wave of active inference is from a 2019 paper from Carl friston so a free energy principle for a particular physics particular here's the joke it Cleaves off particles we call the entity a cognitive particle whether it's rock metronome or like more sophisticated Etc and then it's a physics for those kinds of partitioned systems so that is taking a basian graph and then doing certain partitions on the graph there's other uses for the term partition like you brought up the term from stat Mech so here partition function describes statistical properties of a system in thermodynamic equilibrium so there the free energy is like Gibbs free energy like why do candles burn once they're activated forward like why is time going that way why don't candles like unburn so that's on the thermodynamics then over the last like 00 years all of these Mega developments with information and thermog Dynamics like analogies between them or isomorphisms but also Unity between them like with the land hour limit and how much energy does it take to read or write a bit so it's a kind of info thermodynamic bace that the free energy principle is related to um so are there well-described summary variables that describe systems that are in different kinds of info dnamic equilibrium there are so the simpler case is like a fixed points so that be like the rock is just resting on the ground it's like the belief is not updating then the next step would be like the simple update like this bilard ball hit the rock it moved this much that's like this data point hit the prior it moved that much and then there's all kinds of like steady states that could be like an oscillator so you're going the right area to see how those things are formalized yeah that um so I guess it isn't like it wouldn't be describing because I did read on on YouTube there it was saying the thermodynamic properties of a system like the temperature and volume and uh so if in an in what would those be um parallel to like in an active inference system would it be the aspects like decision making and balancing exploration and learning great question Andrew what would you say to that um I mean yeah it's I'm trying to figure out how how to phrase that well I mean it's you'd want to look at the I would just refer to like the the breakdowns of variational and expected free energy I suppose it's it's it's expected free energy where you're going to see that as I mentioned earlier that epistemic and pragmatic value um differentiation and that that plays out in learning as well like expect like free energy is being minimized during the process of of of learning where rather than you know inference in the context of of active inference like inference is inferring states of the world uh learning is inferring uh sort of model like parameters values right so you're you're going to be like that the agent learns to adjust its own parameter values it's also going to infer Moment by moment States um ex uh the pragmatic and osmic value play into that overall process and and I mean all these things are kind of happening um s mously so so so for me it's it's a little tough for me to like kind of rip out one part of that process and say that one thing is like what's happening um but I'm not sure if I'm answering the question super well or not that's that's I think that was a key piece which is the epistemic and pragmatic value are a way to like reweight the policy prior into the policy post interior so for a kind of thing that's not doing the cognitive task of something like policy updating so if something's drawing from habit that could be like a go to the simplest and fixed habit it's like the rock is always in location one so it's like it's being sampled from location one with 100% again and again it's kind of like a redundant or a tot logical statistical distribution and that is also what comes up is like def defining things as they're measured to be so then rather than the the space of possibly dissipative things you say like well the measurements over this time scale are such that it is returning to an attractor and then what are the Dynamics of that attractor is it like just a marble going to the bottom of a bowl and perturbation brings it back or does the bottom of the bowl adjust that's like learning is there another landscape then all all those kinds of questions you think it'd be helpful to thank you I don't know there's I like the um one of one of the simpler equations from active inference for just like Computing like the final posterior over policies I I just I've always found that interesting and like helpful for like describing to folks lesser familiar with active inference like like your your posterior over policies typically is like um a softmax fun softmax function over negative expected free energy times gamma which is referred to it's like a um Precision over actions and then um minus um I'd rather pull it up actually just so I don't miss speak here the the main point is that it includes expected free energy and variational free energy and habits all of which you know Daniel just mentioned a moment ago and it helps to give a an intuitive sense of like all three of these components are are are going into the agent's Behavior like what action they end up selecting right it's like what expect expected free energy as what it expects to happen and that's where it's uh prior over observations come in so it's that's where its preferences come in and then um variational free energy is just reducing kind of uncertainty in the moment and then habits are like these empirical priers that have been built up over time yeah another kind of big moment in helping think about this was this particular kinds showing continua from like in continua of modeling continuity not as much as like an ontological Claim about any actual system but in terms of a modeling continuity with systems that are being interfaced with differently ranging from kind of like inert unidirectional cause to kind of simpler inert to simpler nonplanning like on through like planning like with other cognitive phenomena like reflexivity meaning creation things like that so the simple ones really do look like this the graphs but this graph on the bottom is way more of like a schematic and in play those graphs might be very large um sh Harsha or Courtney have any question or like different direction to explore I don't at the moment I was just kind of listening to your guys conversation and seeing thank you C from that since I'm not that far along yet cool thank you yeah it's like always like skipping around on on a record here's one this is the basian mechanics paper this is kind of a summary slide thank you shha so this paper this is like docon stiva devel and Maxwell ramstead and others working on the basian mechanics so here's classical physics classical mechanics stationary action that's like why the baseball goes in the parabola then statistical physics says well it's not just like planets and baseballs and gravity it's other certain behaved summary statistics like temperature and pressure that was what we were just looking at with the partition functions um Quantum moves that into the not just the scale that was being explored but also like the imaginary or like axis that's orthagonal so that's why there's the imaginary numbers and the rotation and it's an equivalent physics of Uncertain systems especially interpreted in this kind of scale-free way so it's not just like unique mechanisms related to electrons but those contribute to it and and uh constitute why the variability patterns are the way they are but people have applied Quantum formalisms across different systems even without relying on like some sort of um Reliance on electrons or protons or anything and then kind of in that same frame and generalization and you can generalize Quantum to be statistical physics with no um wave distribution and you could summarize statistical physics to be like that's why the baseballs um moves like a parabola instead of like diffusing basian physics extends that to arbitrary State space so quantum mechanical State spaces maps of quantum mechanical or Stat or classical are certain within certain categories of maps and models so this basing mechanics thread especially a lot of the rigor that Dalton and Chris fields and some others brought in is only in the last like two post again post 2019 prison kind of entry into that fundamental uh level of the interface but it's kind of also threaded going prior but took a lot of a a consolidation and development turn and this is not brought up in the 2022 textbook so it's kind of like the textbook is at this checkpoint where the like the Baseline the so in the textbook it's like what are the math areas and it says linear algebra so matrix multiplication and then dynamical and stochastic systems like probability on dynamical systems that is like a checkpoint and then that's been some of the most interest over the last years was what what is the basis of basy mechanics how does it it matter or differ from the um 2022 textbook Style Andrew anyone can give a thought or a different question yeah sorry Dana what what were you refering to in terms of the linear algebra and the stochastic probability um but in in the textbook so one part is where they sort of uh mention the kinds of maths that are needed like those are good keywords to see like here linear algebra calculus and and series expansion and then also there's the appendices but these appendices are kind of just like the the um the tip of the iceberg uh a is kind of some of the code but there could be a lot of work in Matlab and other languages to make code examples like that's what we're doing in Julia right now appendix B goes through the equations but still it's only showing some and then here's oh c is the code so a must be the background yeah have you ever have you ever watched oliv Oliver Nails uh YouTube put a link I don't put a link I don't recognize or no so here's the four topics linear algebra series approximation like series expansions function approximation variational calculus and stochastic Dynamic so systems that have dynamical nature and probabilistic nature but that's the math that's what but this is also foundational for going further into basing mechanics because in many ways it's just like more um remixes and and kind of like concatenations of these core mathematical motifs there's been [Music] um there's been many people's contributions to making the math more um relatable it's just like it's a it's a it's a great question to an area to kind of run towards and contribute to and explore like Jonathan shock is a math professor he wrote this gray text um however it it may be like you know more or less than someone is looking for but it is really cool even like like like even a P value or or um like a bit there's like a lot of things to return to so it's not like a check it off your list because the basis of this is just like surprise and entropy so then that's like why it's important to return to it many times because it's not like you can just cram it and finish the topic and then just like never return to that node here are 10 short cool cool yeah and then in the textbook group like the work that uh octopus is doing right now over the years people have brought up many questions and thoughts about learning math and about how Math Connects and then octopus is now like more um holding down these meetings and just trying to coordinate more math educational work but something that that's always there always improvable if somebody wants to start like just on any part I think Axel made some models and also mentioned that if people were interested in that he'd be open to collaboration too octopus said that Axel Axel constant um Axel Sorenson I think in the chat oh Axel Sorenson okay cool yeah cool yeah like for the math and stuff so people could see the inter relationship between the terms and things like that yeah on on on the terms uh if people want to follow up that like more formal but natural language so not using equations per se but um going into like terms and in a relationship that's like another dimension with the active inference ontology so that's the like that's the generic system description and then those are the terms that actually apply to the generic system so in certain ways it's simpler than trying to give examples from systems because it could be like oh the action state of the robot well that's like you know the the motor it's like well but then the motor has a sense element and then from the world's perspective it's it is a you know so it gets really specific when you are dealing with specific systems but the core ontology is you don't even need to use these 64 terms it's very small and the kind of Kernel of it is the particular partition and why and how there's a a least action principle on a certain kind of partitioned graph with so all the equations just setting up the path but then that's why these kinds of um nodes and Edge figures come up all the time because that is the setting that's being considered and that's coming more from like the basic mechanics free energy principle side whereas um in in the textbook they actually set a target for active inference which is why in figure 1.2 active inference is is at the middle between that free energy principle basy mechanics and then the bottomup construction of real math and computational models so this is more like applied math and applied computer science because it's like you're making matrices that represent a certain system or certain measurements and then up here is more like the generic it's like like why does gas burn why does wood do this and then here's like this specific this particular engine so then that helps recognize like somebody will make a claim analogously to something like well because candles burn my car will get 200 miles per gallon it's like not necessarily or because metal has some structural Integrity this bridge design is going to be fine so the bottom and the low road is like a very empirical inquiry and building and then here is just some of the more generic elements that are related to physics so that's why this book brought things together structured it in a different way than like was existing because it it um brings in that low road High Road approach which can be like dug into and you can go deeper into low and high road but just the layout of like kind of setting context low in the high road just one way to think about it but a very useful one and then chapter four when they come together and then chapter five which is the most well-studied domain at this time which is mammal Neuroscience so that's why even though there's sections like especially in four that are just like whoa what is happening um and and eventually we'll have more of like generative dials like turn up the math detail turn down the math detail but um in a balanced form going from here's the context then take these two paths here's what it is here's where it's being used it does have those trails um yeah the math the math learning education Courtney thank you I'll copy it in so like octopus can explore it yeah PE people have made many continua um what's kind of fun is there's there's hope for like doing something in a very different way because if you think about a technical concept Like A P value it's hard enough to communicate technically and basian um prior and posterior and updating even before getting into it just if they're learned one by one and through the kind of Tech Tree Learning trees that are how they're currently taught it's going to a very long road with a lot of attrition so I think that's what motivates some of these questions like about math learning that people have um brought up over the years that's a really interesting layer to pull back to because like computers are used without math um needed by the user so as that Continuum and that's doc starts to materialize like what and how are people learning what would be of epistemic and pragmatic value for them all those kinds of questions then San jev's textbook the fundamentals uh which he has finished the first volume of it's um two volumes or this is just what I understand now first volume is related to active inference so that's kind of coming at it from the low road largely second volume is free energy principle phasing mechanics however the the first volume is extremely comprehensive gives a mega synthesis that's that will be um comprehensive enough for like very serious extended study and also use in language models Etc um so that will be very nice on the low road and then I think that's why is because it gives a more time for codification and synthesis on the high road Elements which I mean even if there is as uh minim I mean that's the whole thing with Maxwell's equation and Newton's equation is sometimes it IT projects down to these Expressions that are just so small yeah I mean there's a lot of a lot of there's a lot to go into here I suppose I suppose that um actually like typically in an act of inference you're you're dealing with equilibrium States but you're always working it's always working to minimize the free energy so like it's it's non equilibrium yes importantly it can include that situation so like you could have a um a cup of coffee in the room and they're at thermal equilibrium but then you have non-equilibrium thermodynamics which is like Heating and Cooling and more phenomena if it was like a huge temperature difference or something so similarly it's good to check like yes I'm taking in this one data point I'm updating the prior here it is like and have a kind of um like it's at equilibrium like the temperature the thermometer setting 70 my estimate is 70 it should stay at 70 but then you can explore all of these situations where that is relaxed yes so you have like a Colman alter which is like kind of a sliding window memory then if you add it as just one thermometer measurement it's going to jerk around and just reflect the measurements or if you had really like too heavy it would just start taking long run average so then that's like a basian question like what should the window and the forgetting be on the common filter and then the whole thing with the particular partition is like now imagine if you were fitting that on the flows of all four of these so for each variable they're all kind of undergoing that flow-based unfolding which could be static or it could be noisy or it could be um bodal it's like trapped here and then it's trapped here and then like what is the structure of all those variables it's not just for if you want to do something more than a thermometer abstraction but this is like the kernel that describes the particular partition and why and how that's coming from um or why and how you can have a path of least action on that flow which is what is set up here and and improved and clarified also in but but this will always come up it's like there's the four states sense action internal and external so fun and different letters depending on different people's use and so on um and then what their their time development is plausibly a function of and that's reflected by the connections between the nodes um like that is what this is showing which is that there isn't a direct Edge between internal and external that's tological because it's what defines the interface if there was a direct Edge you're just talking about a different situation but these don't necessarily map on so again none of this is a claim about how things are structured in the world it's like if you are using a base graph anything other than a fully connected graph so if you have everything's connected to every node then there's no partition it's just like one connected Mass um if there are pieces that aren't connected to each other then this situation comes into play that that's interesting yeah totally and it's like a it's like a separate sort of um little subgraph yeah exactly like the the the um this is the niche and this is our Roomba you know act the motors again just simplifying the motors and the internal States and the temperature reading it's like we we have partitioned the Roomba from the niche but if there was like some sort of wire between them inter is then it wouldn't be internal States so it's like that's the whole chapter six Plus+ plus question how do you make these kinds of maps for cognitive systems and and that helps sometimes to clarify like the the epistemic status of these Maps because like why is this reflected that way it's like because it was constructed that way as the map but they they don't like just taking alone support or reject some kind of way that the world is but making them within a certain um type brings the generality of cognitive modeling and then that's kind of just like the the how specific will you make the model like is the engine just one chamber and I just like I'm just looking at heat across an engine overall or do go and model Heat at like the components or would do you model heat transfer with this but those are all empirical and the the person of the team building that just addresses those as they actually happen how many chambers should we model this engine is having it's like why are we building the model who's it for what questions are being asked that's what chapter six goes into um Jeff also this morning and just like we're that was fun it was about the inspiration in the dream but it was it it was fun and and uh but the the pendulum is going to give more that's going to come across more as like a uh that's more like solving an exercise in a physics class if it's like we have an oscillator with two states it's oscillating due to this write the generative model that does this that's kind of that I mean it has like a kind of homework style feel not that there'd be only one answer but there would be a Continuum of problems from like there's only one answer as stated to there's multiple ways that you could have done that on through I mean it was just like where's time time perception but then those those are those can you can still play with it and and provide and take different thoughts on it even if it's not and then it just like oh well it's not about generating one specific model but if it was um if we're a startup and it the the financial um incentive was to make the thermometer with this or that property then it' be again just more of an engineering question also how it kind of revisits like chapter 4 shows discreet and continuous in one chapter like what's shared about them and then seven and eight revisit it that's like just showing even from the beginning like it's the situation choice to model even how time is okay okay yeah I'll I'll put this in math learning resources you you can add things um Adam or anyone like and if you're not sure yeah that's collaborative fun it's adding questions I mean the hundreds of questions that have been asked did not come from nowhere and somebody with many questions could contribute many questions and someone who goes through many resources could just say here's what this one I just found it this way no one's going to think it's like a total final judgment just this was useful for me learning this from here at least it helps show what different kinds of activity and different like practices can mean and it it just helps show examples oh yeah these are ways to there's no um death gap between like not knowing linear algebra and slightly understanding it but then that that comes down to everyone's specific situation how much time and what kind of thing and like yeah graphs like the the learnable loop that we this most recent blog post where kobus is able from RX and fur for the a drone model for this to be automatically generated it's it's an incredible game Cher it's it's amazing work it like Leap Frogs what visualization capacity for example exists in um python or um matlb not not because it's like a new visualization package it's just like you would be kind of starting on Square zero like okay I have a BAS graph represented in mat labra and python but I'm I going to just do like spread them all out or how is this even going to work and for it to Output like publication ready and with a verified Integrity from how it's actually specified not being like this extensive PowerPoint um just kind of offchain moves which is doing doing one's best in the absence of this but I mean this is like wow so what what what exactly is that that graph figure there yeah great question so there's there um and like continue to dig and and explore the papers in live stream where this is explored more but basically briefly there's there's two primary kinds of graphs that come into play in um active inference the first one is a basian graph so in that situation the no are random variables and the edges res um resemble like causal influence so that's like causal modeling um all those kinds of like basian models from J Pearl and others nodes are variables edges are like functions that relate variables their Maps it turns out over the last like 10 years being applied more to to like Bert devise and Carl friston but the fundamentals kind of lying in the pre previous decades every base graph has also this dual representation as what's called a forny graph yeah porny graphs basically the nodes and the edges have the the opposite character so it's like whereas in if in a base graph you could have one node with like 50 connections because it could just be influenced by 50 variables um whereas The Edge can only have two nodes it's connected to the forny graph kind of inverts that where the nodes are like um edges in the base graph and vice versa and it there's there's like a dual relationship you can kind of represent them both ways and and go between those two but it turns out that in certain forny graphs you can take the the basian causal graph um you know probability of rain probability of that here's the observation so that's the kind of like systems thinking how are things connected to each other and then there's this rendering that RX and fur does under the hood to another structure that's equivalent and does the same calculation but it does it in this way that can be like scheduled whereas on a Bas graph there'd be a lot of Dees of freedom in like how do you properly update this and percolate messages and the factor graph gives a much stronger way to do that this is a technical detail but it's like what is making the RX infer method more oriented towards the scaling and the engineering type efficacy because it's it's a super strong engine under the hood for running these 40 Factor graphs but also enables this more basy and semantic layer possibly for people to interact with it but to see it like we've seen it before um and flip it also into the version that actually relates to how like the different things connect so just like new level of of um ability to to look into the the kind of base computational layers of these models in in but not by just making like custom visualizations like here's one some other large one like for and also time varying autoaggressive this highlights which is so um Wise by kobis like this isn't just about cybernetics this is just doing basy and statistics like on time series which could be like temperature climate markets Etc so it doesn't always need to relate to more questions about like embodiment or communication amongst agents they can just be like this is the graph and then that's just a regression so it's not a theory about agents anything specific or any specific type of agent per se I mean everything of course I'm just saying saying informal takes but the fact that Kus is showing no you can use RX and fur to do these just linear aggression Auto regression no no perception action here okay um well whatever uh I also I updated the calendar event so that it should say a little more clearly like which piece is being discussed but um yeah we'll just continue on so any last comment or otherwise thanks thanks lot cool appreciate it farewell see you later